

Predictive Forecasting of Care Load and Placement Demand for Unaccompanied Children Using Time-Series Analysis and Machine Learning

Ayush Sharma

Research paper based on the Predictive Forecasting of Care Load & Placement Demand project

Abstract

Operational planning for the care of unaccompanied children depends on understanding how the number of children in care changes over time. This paper presents a forecasting workflow for estimating the care load reported for the U.S. Department of Health and Human Services (HHS) Unaccompanied Alien Children Program. The project uses a historical operational dataset containing dates, children apprehended and placed in Customs and Border Protection (CBP) custody, children in CBP custody, children transferred out of CBP custody, children in HHS care, and children discharged from HHS care. The target variable is “Children in HHS Care.” The workflow includes data cleaning, chronological sorting, exploratory data analysis, temporal feature engineering, baseline forecasting, statistical time-series models, and machine-learning regression models. The project compares a naïve baseline, ARIMA, SARIMA, Holt–Winters, Random Forest, and Gradient Boosting using mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE). The saved comparison notebook reports Gradient Boosting as the strongest candidate, with MAE 58.79, RMSE 79.32, and MAPE 2.70%, followed by Random Forest with MAE 67.36, RMSE 89.08, and MAPE 3.12%. However, the project artifacts also reveal irregular date coverage and inconsistencies between some individual notebook metrics and the consolidated comparison table. Therefore, the reported results should be interpreted as preliminary rather than deployment-ready evidence. The study demonstrates a practical forecasting pipeline and identifies data continuity, reproducible evaluation, and operational validation as essential next steps before forecasts are used for resource planning.

Keywords: time-series forecasting, care demand, HHS, unaccompanied children, Random Forest, Gradient Boosting, MAE, RMSE, MAPE

1. Introduction

Forecasting operational demand is important for organizations that must plan staffing, accommodation, services, and other resources under changing conditions. In the context of the HHS Unaccompanied Alien Children Program, the number of children in care is a practical measure of the current care load. A forecast of this quantity may help analysts understand near-term changes and support planning discussions. Such forecasts should support, not replace, operational judgement and the needs-based decisions made by responsible agencies.

Time-series forecasting can use previous observations of a target to estimate future values. Statistical methods such as ARIMA and seasonal ARIMA (SARIMA) model temporal structure directly, while machine-learning regressors can combine lagged target values, rolling statistics, calendar variables, and operational indicators. Their relative performance depends on data quality, temporal regularity, forecast horizon, and validation design. Comparing multiple model families against a simple baseline is therefore an important part of a defensible forecasting workflow.

This project develops an end-to-end pipeline around the target “Children in HHS Care.” It includes data preprocessing, exploratory analysis, feature engineering, model fitting, metric comparison, future forecasting, and a Streamlit dashboard. This paper documents that workflow and reports the values available in the project notebooks, while explicitly noting limitations that need to be addressed before interpreting the results as reliable real-world forecasts.

1.1 Objectives

- Prepare the historical operational data for time-series analysis.
- Explore temporal patterns and relationships among operational variables.

- Create lag-based, rolling-window, and calendar features for machine-learning models.
- Compare baseline, statistical, and machine-learning forecasting approaches using MAE, RMSE, and MAPE.
- Summarize the strongest preliminary model and identify the checks required before operational use.

2. Related Work and Background

Forecasting literature commonly recommends beginning with simple benchmark methods and evaluating more complex models against them. Hyndman and Athanasopoulos describe a forecasting workflow that includes visualization, time-series decomposition where appropriate, benchmark methods, residual assessment, and evaluation on observations not used to fit a model [1]. ARIMA models represent temporal dependence through autoregressive and moving-average terms, with differencing used when needed to address non-stationarity [2]. Seasonal ARIMA extends this approach to recurring seasonal structure.

Machine-learning methods offer a different approach. Random Forest combines predictions from many decision trees and can capture nonlinear relationships and interactions [3]. Gradient Boosting builds an additive sequence of weak learners, with each stage improving on errors made by the current ensemble [4]. In forecasting applications, these models generally require careful construction of features from information that would genuinely be available at prediction time. Lag variables and rolling summaries are common choices, but they must be generated without allowing future target values to leak into the training data.

The present project applies these model families to a public-sector care-load forecasting problem. Its practical contribution is the implementation and comparison of several approaches in one pipeline, accompanied by an interactive dashboard. The work should be considered an applied project study rather than a claim of a new forecasting algorithm. A stronger research evaluation would require a fully reproducible temporal validation design, consistent metric calculation, and comparison with additional established benchmarks.

3. Dataset and Preprocessing

The project README identifies a historical dataset concerning the HHS Unaccompanied Alien Children Program. The exploratory-analysis notebook reports 720 records and six columns in the cleaned dataset. The variables documented in the project are summarized in Table 1. The target is the number of children in HHS care; the remaining operational measures may help contextualize changes in care load or provide candidate predictors where their values are available at forecast time.

Variable	Description / role
Date	Observation date; used to order records chronologically.
Children apprehended and placed in CBP custody	Operational flow indicator.
Children in CBP custody	Reported number of children in CBP custody.
Children transferred out of CBP custody	Operational transfer indicator.
Children in HHS Care	Target variable representing the current care load.
Children discharged from HHS Care	Operational discharge indicator.

The documented preprocessing steps include loading the raw data, converting dates to datetime values, removing commas from numeric values, checking missing values and duplicates, sorting observations by date, and preparing the data for time-series analysis. The feature-engineering notebook adds calendar features and target-history features such as `HHS_Lag_1`, `HHS_Lag_7`, `HHS_Lag_14`, `Rolling_Mean_7`, `Rolling_Mean_14`, and `Rolling_STD_7`. A `Net_Pressure` feature is also listed in the engineered data.

An important data-quality issue is visible in the notebook outputs. The feature-engineering notebook reports 1,047 expected calendar dates, 706 available dates, and 341 missing dates over the considered span. It also reports a date index without an associated frequency in the statistical-model notebooks. This irregular coverage means that a row lag cannot automatically be interpreted as a one-day lag, and a seven-row rolling window may not represent seven consecutive calendar days. The data should therefore be checked for whether missing dates mean no report was published, a zero value, or an unavailable observation. These possibilities should not be treated as equivalent.

The analysis notebooks report no null values in the original six-column dataset after cleaning, but absence of nulls does not establish complete daily coverage. For future work, the project should explicitly check duplicate dates,

frequency, gaps, and the meaning of missing dates before filling or resampling the series. Any imputation or calendar regularization decision should be documented and tested for sensitivity.

4. Methodology

4.1 Exploratory Data Analysis

The project includes exploratory analysis of data types, descriptive statistics, time-based patterns, missing dates, and relationships among operational variables. The purpose of this stage is to understand the target's scale and variation, identify data-quality problems, and guide feature creation. The notebooks also include an Augmented Dickey–Fuller (ADF) stationarity test. The saved output reports an ADF statistic of approximately -0.942 and a p-value of approximately 0.774 ; at conventional significance levels, this result does not provide evidence to reject the unit-root null hypothesis. This test result should be interpreted with caution because the time index is irregular.

4.2 Feature Engineering

The machine-learning pipeline uses recent values of the target and rolling summaries to represent local level and short-term dynamics. Calendar features include year, month, day, quarter, day of week, and a weekend indicator. The project also includes operational variables and a Net_Pressure feature. Lag and rolling features must be created only from past information, with rolling calculations shifted appropriately where necessary. At inference time, the feature-generation procedure must reproduce the same ordering, missing-value treatment, and column definitions used during training.

4.3 Forecasting Models

Six approaches are listed in the project's model-comparison notebook:

- Naïve baseline: uses a simple historical reference prediction and establishes a benchmark for more complex models.
- ARIMA: models autoregressive and moving-average dynamics with differencing.
- SARIMA: extends ARIMA with seasonal terms; the saved notebook reports a weekly seasonal specification.
- Holt–Winters: uses level, trend, and optional seasonal components through exponential smoothing.
- Random Forest Regressor: combines decision trees to model nonlinear relationships between engineered features and the target.
- Gradient Boosting Regressor: sequentially combines learners to reduce predictive error.

4.4 Evaluation Metrics

Model evaluation uses MAE, RMSE, and MAPE. MAE is the average absolute difference between actual and predicted values; RMSE penalizes larger errors more strongly because errors are squared before averaging; MAPE expresses absolute error as a percentage of actual values and can become unstable when actual values are zero or close to zero. Metrics are meaningful for comparison only when calculated on the same held-out observations, at the same forecast horizon, and with the same preprocessing and metric definitions.

For a rigorous time-series evaluation, training data should precede validation/test data chronologically. Random shuffling is generally inappropriate because it can allow information from later periods to influence model selection. A rolling-origin or expanding-window evaluation would provide a more robust view of performance across multiple forecast origins.

5. Results

Table 2 reproduces the model-comparison values saved in the project's comparison notebook. These are project-reported results, not a newly rerun or independently reconciled benchmark. Lower values indicate smaller errors when the metrics are computed on a common evaluation set.

Model	MAE	RMSE	MAPE
Gradient Boosting	58.79	79.32	2.70%
Random Forest	67.36	89.08	3.12%
Baseline*	235.26	285.70	11.12%
ARIMA*	235.26	285.70	11.12%

SARIMA	643.99	870.78	28.04%
Holt–Winters	843.79	939.78	38.32%

*Metric consistency note: the baseline notebook separately reports MAE 196.09, RMSE 239.35, and MAPE 9.24%, while the comparison table reports 235.26, 285.70, and 11.12%. The ARIMA notebook also contains output that does not align with the consolidated comparison table. These differences must be reconciled by rerunning the evaluation before the table is treated as final evidence.

According to the saved comparison table, Gradient Boosting has the lowest reported error across all three metrics, with Random Forest in second place. This is consistent with the usefulness of recent target lags and rolling summaries in predicting a slowly changing series. The Gradient Boosting notebook reports the largest feature importances for HHS_Lag_1 (approximately 0.747) and Rolling_Mean_7 (approximately 0.171), followed by HHS_Lag_7 (approximately 0.044). These values suggest that the model relies heavily on recent target history. Feature importance indicates how the fitted model uses inputs; it does not establish that a feature causes changes in care load.

The statistical models show substantially higher errors in the saved comparison. However, because the date index is irregular and the comparison notebook contains metric inconsistencies, the ranking should be interpreted as preliminary. A fair comparison requires all methods to predict the same test dates with a consistent horizon and evaluation code.

5.1 Illustrative Future Forecast

The future-forecast notebook contains a 30-step output dated from 22 December 2025 through 20 January 2026. Its saved summary reports a forecast mean of approximately 2,562.83, a minimum of approximately 2,479.80, and a maximum of approximately 2,606.70. These values are reported here only to describe the project artifact. Because the forecast dates are historical relative to the current date and because the input series has irregular date coverage, this output should not be presented as a current forecast or as validated future demand.

6. Discussion

The project demonstrates an applied workflow that connects data preparation, time-series feature engineering, multiple model families, error metrics, and an interactive dashboard. The reported machine-learning results are promising within the existing experiment: Gradient Boosting and Random Forest outperform the listed statistical methods in the consolidated comparison table. The importance assigned to recent lags also aligns with the intuitive persistence of a stock-like measure such as the number of children currently in care.

Nevertheless, the strongest conclusion is methodological rather than operational. Missing calendar dates and conflicting metric outputs weaken confidence in the numerical ranking. A model may appear highly accurate if the test period is easy to predict, if features contain information unavailable at forecast time, or if different models are evaluated on different dates. The study therefore does not claim that the current scores establish readiness for agency decision-making.

6.1 Limitations

- Irregular date coverage: the project reports 341 missing dates across a 1,047-date span.
- Metric inconsistency: individual model notebooks and the consolidated comparison table disagree for at least the baseline and ARIMA results.
- Validation details need stronger documentation: the exact shared test window, forecast horizon, and model-selection procedure should be made explicit.
- Potential feature availability: operational predictors should only be used when their values would be known at the time the forecast is issued.
- Limited scope of model comparison: additional naïve seasonal benchmarks and rolling-origin evaluation would help assess robustness.
- The forecast output is not current and should not be interpreted as a present-day estimate.

7. Recommendations and Future Work

- Audit date coverage and duplicate dates, then document why observations are missing before selecting a resampling or imputation strategy.
- Rebuild a consistent chronological train/validation/test split and evaluate every model on exactly the same test dates.
- Centralize metric calculation in one reusable function and regenerate the comparison table directly from the predictions.
- Prevent leakage by ensuring lagged and rolling features use past values only; verify predictor availability at each forecast origin.
- Add a last-value baseline and a seasonal-naïve baseline, then use rolling-origin evaluation to measure stability across time.
- Report uncertainty intervals where supported and communicate forecasts as planning aids rather than guaranteed outcomes.
- Update the Streamlit dashboard to display the data freshness date, evaluation period, selected model, error metrics, and limitations.

8. Conclusion

This paper presented a practical forecasting pipeline for the HHS care load of unaccompanied children, using “Children in HHS Care” as the target. The project includes preprocessing, exploratory analysis, lag and rolling-window features, statistical forecasting, machine-learning regression, model comparison, and a Streamlit dashboard. In the saved comparison table, Gradient Boosting achieved the lowest reported errors (MAE 58.79, RMSE 79.32, MAPE 2.70%), followed by Random Forest (MAE 67.36, RMSE 89.08, MAPE 3.12%). These findings are preliminary because date gaps and inconsistent metric outputs remain unresolved. The most important next step is to regularize and audit the time index, rerun all models under a shared chronological evaluation protocol, and regenerate the results. With those checks in place, the project can become a more dependable analytical aid for understanding potential changes in care demand.

References

- [1] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021. Available: <https://otexts.com/fpp3/>
- [2] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. Hoboken, NJ, USA: Wiley, 2015.
- [3] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.
- [4] J. H. Friedman, “Greedy function approximation: A gradient boosting machine,” *The Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001, doi: 10.1214/aos/1013203451.
- [5] U.S. Department of Health and Human Services, Office of Refugee Resettlement, Unaccompanied Alien Children Program. Program data and definitions should be verified against the source dataset accompanying the project.
- [6] A. Sharma, “Predictive Forecasting of Care Load & Placement Demand,” project notebooks, processed datasets, model artifacts, and README supplied with the project archive, 2026.

Reproducibility Note

This paper was drafted from the project README, processed datasets, and saved Jupyter notebook outputs in the supplied project archive. Model metrics are transcribed from those artifacts; the models were not fully rerun for this document. Before final academic submission, confirm the required citation style, institutional template, author affiliation, and whether the source dataset requires a formal URL and access date.